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FORCAST grade level: 0

Gunning Fog Index: 0

Dale-Chall Index: 0

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NGỮ PHÁP & CHÍNH TẢ — 1 LỖI PHÁT HIỆN

Xếp loại: **B**

Factors Influencing the Effectiveness of the Flipped Classroom Approach in Teacher Education Programs at Universities

Abstract

Quantitative methods were employed to identify the key factors influencing the effectiveness of the Flipped Classroom approach (FCM) in English language teacher education programs at universities. Survey data were collected from students, lecturers, and administrators across three universities and analyzed using exploratory factor analysis and regression techniques. The results indicate that participants reported significantly higher levels of satisfaction and acceptance toward the FCM compared to traditional teaching methods. The findings indicate broad support for the FC approach and confirm its positive association with student learning outcomes. Regression analysis identified several strong influencing factors, including lecturers' responsibility and competence in pedagogical innovation, the level of technology integration in teaching, competency-based assessment practices, and the promotion of student autonomy. Additionally, institutional conditions such as physical facilities, technological infrastructure, and the availability and openness of digital learning resources were found to play an important role. Effective instructional design and lecturer professional development further strengthened implementation outcomes. However, large class sizes, limited learning resources, and workload-related constraints remained important challenges.

Keywords: Flipped Classroom teaching method, Flipped Classroom, Learning Effectiveness, Barriers, Student Engagement

1. Introduction

Vietnamese higher education is undergoing substantial reform as universities seek to strengthen competency-based learning and accelerate digital transformation. Within this context, the flipped classroom (FC) has gained increasing attention as a learner-centered approach that promotes active engagement and independent learning. In a flipped classroom, students engage with core content before class, allowing classroom time to be used for discussion, collaboration, and higher-order learning activities. Classroom time is optimized to focus on higher-order thinking activities such as discussion, collaboration, critical thinking, and problem-solving (King, 1993; Bergmann & Sams, 2012; Bishop & Verleger, 2013; Freeman et al., 2014). Since its initial dissemination in K12 education in the United States (Bergmann & Sams, 2009; Lage & Platt, 2000), this model has rapidly expanded and been applied in higher education worldwide.

The Flipped Classroom teaching method is not only limited to improving academic achievement and student satisfaction (Cho Naing et al., 2023) but also aims to foster critical thinking, promote active learning, and enhance students' self-directed learning (Dzaiy et al., 2024). However, despite the recognized benefits, some studies have indicated that implementing this model may increase learning pressure and workload for students, while also requiring high levels of instructional design competence from lecturers (Tang et al., 2017). Notably, in the context of Vietnamese higher education, particularly in teacher education programs, empirical studies on the factors affecting the effectiveness of the flipped classroom model remain limited. Most previous studies have primarily focused on comparing the effectiveness of this approach with traditional teaching methods, without fully analyzing the predictive factors influencing its effectiveness from the perspectives of relevant stakeholders.

To fill this gap, a quantitative study is necessary to analyze the factors affecting the effectiveness of applying this teaching method in training students in Enlish language teacher education programs at universities. The study employs a survey questionnaire as the primary research instrument and aims to achieve the following objectives: (1) to evaluate the levels of satisfaction and acceptance among lecturers, administrators, and students regarding the application of the flipped classroom teaching method; (2) to identify and analyze key predictive factors affecting the overall effectiveness of this approach through regression modeling; and (3) to assess the impact of the flipped classroom method on students' academic performance.

2. Literature review and theoretical framework

2.1. Flipped classroom in higher education

The flipped classroom (FC) has become one of the most influential instructional approaches in higher education because it shifts the focus of teaching from knowledge transmission to active knowledge construction. Rather than receiving information primarily during class, students engage with instructional materials before class through videos, readings, or other digital resources. Face-to-face sessions are then devoted to discussion, collaborative learning, problem-solving, and other higher-order learning activities. Bishop and Verleger (2013) described the flipped classroom as a combination of computer-based individual instruction outside the classroom and interactive group learning during class, emphasizing that its effectiveness depends not simply on the use of technology but on the pedagogical redesign of learning activities.

A substantial body of research has reported positive outcomes associated with flipped learning in higher education. In their systematic review, O'Flaherty and Phillips (2015) found that the flipped classroom generally increases student engagement, motivation, and satisfaction by encouraging learners to participate actively in classroom activities rather than passively listening to lectures. Similarly, Lo and Hew (2017) concluded that students in flipped classrooms often demonstrate stronger self-directed learning skills because they assume greater responsibility for preparing before class and regulating their own learning processes. This shift from teacher-directed instruction to learner autonomy enables classroom time to be used more effectively for applying knowledge, exchanging ideas, and receiving immediate feedback.

Previous research has consistently linked flipped learning with higher levels of student engagement, improved academic performance, and stronger self-directed learning skills (Missildine et al., 2013; O'Flaherty & Phillips, 2015; Divjak et al., 2022), while also reflecting an inevitable trend in advanced education systems (Tran Thi Ai Luyen & Nguyen Thi Kim Anh, 2024).

Beyond improving engagement and self-regulated learning, recent studies suggest that the flipped classroom contributes to the development of higher-order thinking skills. Collaborative learning activities, peer discussion, and authentic problem-solving encourage students to analyze, evaluate, and apply knowledge in meaningful contexts, thereby strengthening critical thinking and communication skills. Divjak et al. (2022) further argued that advances in digital technologies including learning management systems, interactive multimedia, and artificial intelligence supported learning environments have expanded opportunities for personalized learning and continuous feedback, making flipped learning increasingly adaptable across diverse higher education contexts.

Despite these documented benefits, researchers have also identified several challenges that influence the effectiveness of flipped classroom implementation. Students may experience increased workload due to the need for consistent pre-class preparation, while lecturers are required to invest considerable time in instructional design and the development of digital learning resources. In addition, institutional factors such as technological infrastructure, access to learning materials, and organizational support play a critical role in sustaining successful implementation. Consequently, current research increasingly views the effectiveness of flipped learning as the product of interactions among pedagogical design, technological readiness, lecturer competence, and learner engagement rather than as the result of technology adoption alone. This perspective provides an important foundation for examining the factors that predict the effectiveness of flipped classroom implementation in teacher education.

In addition, the development of modern digital technologies such as artificial intelligence (AI), virtual reality (VR/AR), and learning management systems (LMS) has created favorable conditions for implementing this model in a more personalized and in-depth manner (Divjak et al., 2022; Nguyen Kim Dao et al., 2023), thereby effectively meeting societal demands for a high-quality workforce trained in close alignment with practical applications and learning outcome standards (Pham Dao Tien, 2025).

2.2. Flipped classroom in teacher education

In teacher education, the application of this Flipped Classroom teaching method holds particular significance, as this field requires learners not only to master subject-matter knowledge but also to develop pedagogical competencies, critical thinking, and the ability to adapt flexibly to an increasingly modern working environment. Previous studies indicate that the flipped classroom is particularly suitable for teacher education because it provides opportunities for prospective teachers to integrate pedagogical knowledge with collaborative learning, reflective practice, and authentic classroom problem-solving (Missildine et al., 2013; Herro et al., 2017; Berić-Stojsić et al., 2020).

The application of the flipped classroom is particularly relevant to teacher education because prospective teachers are expected not only to acquire disciplinary knowledge but also to develop pedagogical competence, reflective practice, collaborative skills, and the ability to integrate digital technologies into future teaching. By transferring direct instruction outside the classroom, flipped learning creates opportunities for student teachers to participate in authentic learning experiences such as lesson planning, microteaching, collaborative problem-solving, and analysis of classroom scenarios. These activities promote the integration of theoretical knowledge with professional practice and support the development of competencies required for contemporary teaching.

Previous studies have reported encouraging evidence regarding the use of flipped learning in teacher education. Missildine et al. (2013) found that students in flipped learning environments demonstrated higher levels of academic achievement and learning satisfaction than those receiving traditional instruction. Herro et al. (2017) further suggested that collaborative activities within flipped classrooms enhance creativity, reflective thinking, and professional communication by encouraging learners to solve authentic educational problems collectively. Similarly, Berić-Stojsić et al. (2020) reported that flipped learning supports the development of collaborative learning communities in which students actively construct knowledge through peer interaction and shared responsibility. Låg and Sæle (2019), in their review of higher education studies, also concluded that the greatest educational benefits of flipped classrooms emerge when pre-class preparation is closely aligned with interactive classroom activities that require critical thinking and collaborative knowledge construction.

Although existing studies consistently support the educational value of flipped learning, several research gaps remain. First, most studies have focused primarily on comparing flipped and traditional instruction in terms of academic performance or student satisfaction, while relatively few have examined the institutional and pedagogical factors that determine successful implementation. Second, evidence from teacher education, particularly in developing countries, remains limited despite the growing emphasis on digital transformation and competency-based education. Finally, previous research has often investigated student perceptions alone, with less attention paid to the perspectives of lecturers and educational administrators. Addressing these gaps requires a more comprehensive examination of the factors that influence the effectiveness of flipped classroom implementation from multiple stakeholder perspectives.

Current studies mainly compare flipped and traditional classrooms, whereas little attention has been paid to identifying the institutional and pedagogical predictors of flipped classroom effectiveness in Vietnamese teacher education.

2.3. Theoretical framework

This study is grounded in Constructivist Learning Theory (Vygotsky, 1978) and the Technological Pedagogical Content Knowledge (TPACK) framework (Mishra & Koehler, 2006). These complementary perspectives provide the theoretical foundation for explaining how pedagogical design, technology integration, and lecturer competence jointly influence the effectiveness of flipped classroom implementation.

Constructivist Learning Theory argues that knowledge is actively constructed through learners' interaction with learning tasks, peers, and instructors rather than passively transmitted by teachers. Learning therefore becomes most effective when students actively participate in meaningful activities, engage in discussion, solve authentic problems, and reflect on their experiences. These principles closely align with the pedagogical philosophy of the flipped classroom, where students acquire foundational knowledge before class and use classroom time for collaborative inquiry, critical thinking, and knowledge application. From a constructivist perspective, learner engagement, interaction, openness to discussion, and collaborative learning are essential mechanisms through which flipped classrooms improve learning outcomes.

While constructivism explains why active learning promotes meaningful learning, the TPACK framework explains how lecturers can successfully implement such learning environments. According to Mishra and Koehler (2006), effective technology-enhanced teaching requires the integration of three interrelated domains: content knowledge, pedagogical knowledge, and technological knowledge. Within flipped classrooms, lecturers are responsible not only for developing digital learning materials but also for designing meaningful learning activities, facilitating interaction, monitoring learning progress, and providing timely feedback. Consequently, lecturer responsibility, technological infrastructure, digital learning resources, and instructional design become essential determinants of successful flipped classroom implementation.

Drawing upon these theoretical perspectives and previous empirical studies, this study proposes that the effectiveness of flipped classroom implementation is influenced by the interaction between technological support and pedagogical practice. Technological infrastructure and digital learning resources create conditions for flexible learning, while lecturer responsibility and instructional competence determine how effectively these resources are transformed into meaningful learning experiences. Improved instructional implementation is expected to enhance students' engagement, satisfaction, and ultimately their learning outcomes. Accordingly, the conceptual framework guiding this study positions technological infrastructure, learning resources, lecturer responsibility, active learning, openness to learning, and collaborative learning as key predictors of flipped classroom effectiveness.

Figure. Proposed conceptual framework

Technology Infrastructure

↓

Digital Learning Resources

↓

Lecturer Responsibility and Instructional Design (TPACK)

↓

Flipped Classroom Effectiveness

↓

Student Engagement and Satisfaction

↓

Learning Outcomes

3. Research methods

3.1 Research design

The study was conducted using a descriptive, analytical quantitative research design to address the research objectives. The sample size was determined based on the formula of Hair et al. (1998), n = 5 × number of items, with a stratified convenience random sampling method to ensure representativeness in terms of regions, institutional scale, and teaching conditions. Quantitative data from survey questionnaires of 210 lecturers and administrators and 665 students from three universities (coded as U01, U02, and U03) were used as the core for statistical analyses, including: Descriptive statistics: applied to all data, including Educational Studies course scores under both traditional and flipped methods (Pre-classroom, In-classroom, Post- classroom); observed variables in the survey measuring satisfaction levels of lecturers, administrators and students; Independent T–Test was used to test the independence of input quality and output results between classes taught using traditional and flipped teaching methods; Reliability testing method of scales (Cronbach’s

Alpha) was used to assess the reliability of variable groups in exploratory factor analysis (EFA); Linear regression estimation method was used to construct a regression function reflecting the degree of influence of variable groups on satisfaction in applying the flipped method in teaching Educational Studies courses.

Qualitative opinions from open-ended questions were analyzed using content analysis to support and enrich the meaning of quantitative results. The combination of quantitative and qualitative data enabled both statistical analysis of key relationships and contextual interpretation of participant experiences.

3.2 Survey sample

The survey subjects included all lecturers teaching pedagogical specialized courses and students who had completed at least one specialized course. The research sample was randomly selected, ensuring representativeness across various majors such as Mathematics Education, Chemistry Education, Primary Education, and Early Childhood Education,... The sample was selected using a convenience sampling method. Two questionnaires were specifically designed:

Lecturer sample: 210 lecturers directly teaching the courses.

Student sample: 665 students from the second to the fourth year studying at three universities.

3.3 Research instruments

The study employed two questionnaires specifically developed for lecturers and students, based on a 5-point Likert scale (from 1- very low to 5 - very high). The questionnaires were developed through a three-step process: (1) Literature review of scales assessing active teaching methods; (2) Consultation with experts in the field of education; (3) Pilot testing with a group of 5 lecturers and 30 students to adjust language and structure. The questionnaire structure includes: (a) Opinions on traditional methods, (b) opinions on teaching methods based on the Flipped Classroom model, (c) opinions on the Educational Studies course.

The questionnaires were tested for reliability using Cronbach's Alpha coefficient, with all variable groups achieving values > 0.6, ensuring high internal consistency and reliability for analysis. Specifically, the analysis results show that most scales achieved good reliability, with Alpha coefficients ranging from approximately 0.662 to 0.950, indicating that the observed variables have appropriate internal correlations for inclusion in subsequent analysis steps.

3.4 Data collection and analysis

3.4.1 Data collection

The study was conducted over two semesters of the 2024–2025 academic year, including the following stages: instrument design, pilot survey, data collection, and data analysis.

Both secondary and primary data were included. Secondary data were collected from documents, training programs, course syllabi, and general information of the surveyed universities as well as the affiliated institutions of the research team. Primary data were collected through three main methods. First, the questionnaire was designed with both closed-ended and open-ended questions to measure satisfaction and acceptance of traditional teaching methods, satisfaction and acceptance of the flipped classroom model, as well as perceptions of how well the innovation in teaching the Educational Studies course meets current social demands. The questionnaire was distributed in both printed form and online via Google Forms. Second, semi-structured interviews were conducted with a number of students and lecturers to gain deeper insights into feedback, experiences, and suggestions for improving the model. Third, classroom observations were carried out during experimental flipped teaching sessions to record teaching learning interactions authentically.

3.4.2 Data analysis

Data were analyzed using SPSS 26.0 software and Microsoft Excel. Statistical methods included descriptive statistics, comparative statistics, Cronbach’s Alpha reliability testing, and linear regression estimation. The analytical approach combined regression analysis and expert consultation, with a 5-point Likert scale used in the questionnaires.

4. Results

4.1. Survey and experimental results

4.1.1. Teaching method using the flipped classroom model and learning tools

Pre-class phase: Students engage in self-study at home, completing preparatory tasks through materials provided by the lecturer. Students take ownership of knowledge by actively watching theoretical lecture videos, reading learning materials, answering guiding questions, taking notes on content, and recording questions or unresolved knowledge, ensuring that before coming to class they already have a certain level of understanding of the lesson and their own viewpoints on issues to be addressed (O’Flaherty et al., 2015; Cho Naing et al., 2023; Nguyen Thanh Nga, 2025).

In-class phase: Students interact through activities such as group discussions to solve real-world problems and practice handling pedagogical situations, helping them experience knowledge through simulated scenarios and role-playing. From there, they propose solutions or creative ideas and discuss issues that remain unresolved, as well as explore concepts at a deeper level (Strayer, 2012). The teacher plays the role of supporter and facilitator, without re-delivering theoretical content (Cho Naing et al., 2023).

Post-class phase: Students synthesize the knowledge learned, write reflections, draw lessons, and complete exercises applying knowledge to real-life contexts. Lecturers guide students to explore deeper knowledge (Nguyen Thanh Nga, 2025).

4.1.2. Factors affecting the level of satisfaction and acceptance of the teaching method using the flipped classroom model

The analysis process begins with evaluating the reliability and validity of measurement instruments. The results show that most scales achieve good reliability, with Cronbach's Alpha coefficients ranging from approximately 0.662 to 0.950 (all ≥ 0.6), and item-total correlation coefficients ranging from 0.364 to 0.931 (all > 0.3), indicating that the observed variables have appropriate internal consistency to be included in subsequent analyses. Details are presented in Table 1 and Table 2.

Table 1. Cronbach's Alpha coefficients of observed variables of the teaching method based on the flipped classroom model (student survey) Sample size (n = 665) Source: SPSS output

Table 2. Cronbach's Alpha coefficients of observed variables of the teaching method based on the Flipped Classroom model (Survey of lecturers and administrators) Sample size (n = 210) Source: Results from SPSS

Following the reliability analysis, exploratory factor analysis (EFA) was conducted to identify the underlying factor structure and reduce dimensionality among the observed variables. The test results show that:• The EFA results confirmed a stable factor structure and supported the construct validity of the measurement model. Details are presented in Table 3 and Table 4.• KMO = 0.967 (students) and 0.826 (lecturers and administrators) for the group of independent variables in PPDN, indicating that the data are suitable and satisfy the condition 0.5 < KMO < 1, and Bartlett's Test is statistically significant (Sig. < 0.001) for factor analysis. Details are shown in Table 5.

Table 3. Rotated factor matrix (PPDN–SV) Source: Results from SPSS

Table 4. Rotated factor matrix (PPDN–GV & CBQL) Source: Results from SPSS

Table 5. KMO and Bartlett's Test for independent variables Source: Results from SPSS.

The total variance explained reaches 68.446% (students) and 81.444% (lecturers & administrators), indicating that the extracted factors explain more than half of the data variation. Details are presented in Table 6 and Table 7.

Table 6. Total variance explained for independent variables (PPDN–SV) Source: Results from SPSS

Table 7. Total variance explained for independent variables (PPDN–GV & CBQL) Source: Results from SPSS

A separate regression model examined student perceptions of the flipped classroom method. Technological infrastructure, competency-based trends, lecturer responsibility, and other pedagogical dimensions emerged as significant predictors. The full results are presented in Table 8.

Table 8. Regression results for flipped classroom (Student survey)

A more comprehensive model assessed lecturer and administrator evaluations of the flipped classroom. After removing the non-significant predictor F7, six factors remained significant, with lecturer responsibility again emerging as the strongest predictor. The full results appear in Table 9.

Table 9. Regression results for flipped classroom (Lecturer/admin survey)

The differences in results across the three stages were examined to evaluate the impact of the Flipped Classroom model. Both experimental groups (Flipped THG-K14 and THF-K14) showed higher learning outcomes compared to the control groups (Traditional THB-K14 and THC-K14). In the Pre-class activity, the mean score of the traditional group was 7.16, while the flipped group achieved 8.78. For the In-class activity, the mean score of the traditional group was 7.33, whereas the flipped group reached 9.73. The most significant difference was observed in the Post-class activity, where the traditional group only achieved a mean score of 7.53, while the flipped group reached 9.87.

In addition, the standard deviation of the flipped group across all three criteria was lower than that of the traditional group, indicating higher stability and consistency in students' learning outcomes. These results suggest that students in flipped classrooms tend to participate more actively in learning activities across all three stages of the learning process. Details are presented in Table.

Table 10. Statistical results – comparison by class

5. Discussion

5.1. Interpretation of findings

The findings demonstrate that the flipped classroom is an effective instructional approach for Educational Studies in teacher education. Both students and lecturers expressed positive perceptions of the model, while the quasi-experimental results showed that students in flipped classrooms consistently achieved higher learning outcomes than those in traditional classes across the pre-class, in-class, and post-class learning phases. The largest differences were observed during in-class and post-class activities, suggesting that the flipped classroom is particularly effective in supporting higher-order learning processes rather than simply improving knowledge acquisition.

Among the factors examined, lecturer responsibility emerged as one of the strongest predictors of flipped classroom effectiveness across both stakeholder groups. This finding suggests that the success of flipped learning depends less on the availability of technology itself than on lecturers' ability to design meaningful learning experiences, organize collaborative classroom activities, provide timely feedback, and guide students throughout the learning process. Technology functions as an enabling condition, but its educational value is determined by how effectively it is integrated into pedagogical practice.

The results also highlight the importance of institutional support. Technological infrastructure and accessible digital learning resources were found to positively influence students' satisfaction and acceptance of flipped learning. These findings indicate that active learning cannot be sustained solely through instructional innovation; rather, it requires an institutional environment that provides reliable digital platforms, appropriate learning resources, and technical support for both lecturers and students.

Although the flipped classroom demonstrated clear educational benefits, several implementation challenges remain. Large class sizes, the time required to prepare digital learning materials, and the increased workload for lecturers may reduce the effectiveness of the model if adequate institutional support is unavailable. Nevertheless, the consistently high levels of student interaction, information sharing, collaborative problem-solving, and classroom engagement observed in this study indicate that these challenges can be mitigated through appropriate instructional design and effective use of educational technology.

5.2. Comparison with previous studies

The present findings are consistent with previous studies reporting that flipped learning enhances student engagement, academic performance, and self-directed learning. Similar to the conclusions of Bishop and Verleger (2013) and O'Flaherty and Phillips (2015), this study confirms that reallocating direct instruction to the pre-class phase creates more opportunities for collaborative learning and higher-order thinking during classroom sessions. The superior learning outcomes achieved by students in the flipped classes further support the evidence synthesized by Lo and Hew (2017), who argued that active classroom participation contributes to deeper conceptual understanding.

The findings also reinforce more recent studies emphasizing the importance of digital technologies in supporting flipped learning. Consistent with Divjak et al. (2022), technological infrastructure and digital learning resources were found to facilitate student engagement and continuous learning beyond the classroom. However, the present study suggests that technology alone is insufficient to ensure successful implementation. Lecturer responsibility remained the strongest predictor of effectiveness, indicating that pedagogical competence continues to play a more decisive role than technological availability.

The results further align with studies conducted in teacher education. The observed improvements in collaborative learning, classroom interaction, and critical thinking are consistent with the findings of Herro et al. (2017), Berić-Stojsić et al. (2020), and Låg and Sæle (2019), who highlighted the contribution of flipped learning to reflective practice and professional competence development among prospective teachers. At the same time, this study extends previous research by examining the effectiveness of flipped learning within the Vietnamese teacher education context and by incorporating the perspectives of students, lecturers, and educational administrators within a single analytical framework. This broader perspective provides a more comprehensive understanding of the institutional and pedagogical conditions required for successful implementation.

5.3. Theoretical implications

The findings provide empirical support for the Constructivist Learning Theory underpinning this study. Students achieved better learning outcomes when classroom time was devoted to discussion, collaboration, problem-solving, and reflection rather than traditional knowledge transmission. These learning activities encouraged learners to construct knowledge actively through interaction with peers and lecturers, reflecting the central principles of constructivist learning.

The results also support the explanatory value of the Technological Pedagogical Content Knowledge (TPACK) framework. The significant effects of lecturer responsibility, technological infrastructure, and digital learning resources demonstrate that successful flipped classroom implementation depends on the effective integration of pedagogy, technology, and disciplinary knowledge. Rather than functioning independently, these elements interact to create learning environments that promote student engagement, satisfaction, and improved academic performance. The study therefore extends the application of TPACK by providing empirical evidence from teacher education in a developing-country context, where institutional readiness and pedagogical competence jointly influence instructional effectiveness.

5.4. Practical implications

The findings have several implications for higher education institutions seeking to implement flipped learning more effectively. First, universities should prioritize professional development programmes that strengthen lecturers' competencies in instructional design, digital pedagogy, and technology integration, as lecturer responsibility was identified as the most influential factor affecting implementation. Second, continued investment in digital infrastructure, learning management systems, and high-quality digital learning resources is essential for supporting flexible and sustainable flipped learning environments.

In addition, institutions should establish shared repositories of teaching materials and encourage collaborative curriculum development to reduce lecturers' workload and improve the consistency of learning resources across courses. Finally, institutional policies should promote learner-centred teaching practices by providing both technological and pedagogical support, particularly in teacher education programmes where future teachers are expected to develop professional competencies alongside disciplinary knowledge. Such coordinated efforts will enable universities to maximise the educational benefits of flipped learning while addressing the practical challenges identified in this study.

6. Conclusion

This study examined the effectiveness of the flipped classroom (FC) model in Educational Studies courses within Vietnamese teacher education programmes by investigating the factors influencing its implementation and evaluating its impact on students' learning outcomes. The findings indicate that the flipped classroom is a feasible and effective instructional approach in this context. Students, lecturers, and educational administrators expressed positive perceptions of the model, while the quasi-experimental results demonstrated that students in flipped classrooms consistently achieved higher learning outcomes than those in traditional classrooms. The results further revealed that lecturer responsibility, technological infrastructure, and digital learning resources were the most influential factors contributing to the effectiveness of flipped classroom implementation. These findings suggest that successful flipped learning depends not only on the availability of technology but also on lecturers' pedagogical competence and institutional support for learner-centred teaching.

From a theoretical perspective, the study provides empirical support for both Constructivist Learning Theory and the Technological Pedagogical Content Knowledge (TPACK) framework. The findings demonstrate that active knowledge construction is enhanced when technology, pedagogy, and disciplinary content are effectively integrated to support collaborative learning, critical thinking, and self-directed learning. By examining the perspectives of students, lecturers, and educational administrators within a single analytical framework, this study extends previous research on flipped learning and contributes new evidence from the context of Vietnamese teacher education, where empirical studies remain relatively limited.

The study also offers several practical implications for higher education institutions. Universities seeking to implement flipped learning should invest not only in digital infrastructure and learning resources but also in continuous professional development that strengthens lecturers' competencies in instructional design and technology integration. Institutional support mechanisms, including shared repositories of digital learning materials and policies encouraging learner-centred pedagogy, are equally important for sustaining the effectiveness of flipped learning in practice.

Several limitations should be acknowledged. The study was conducted at three universities offering teacher education programmes in Vietnam; therefore, the findings may not be fully generalisable to other disciplines or higher education contexts. In addition, the research focused primarily on short-term learning outcomes and participants' perceptions. Future studies should examine the long-term effects of flipped learning on professional competence development, compare its effectiveness across different academic disciplines, and employ longitudinal or mixed-method research designs to obtain a more comprehensive understanding of how institutional conditions and pedagogical practices interact to influence learning outcomes over time.